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Family Support and Elderly Well-being in China: Evidence from the China Health and Retirement Longitudinal Study

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Abstract Rapid demographic change, economic growth, and evolving social norms have put increasing pressure on the traditional system of filial piety and family-based eldercare in China. This paper examines the current role of intergenerational transfers in providing old-age support using nationally representative data from the China Health and Retirement Longitudinal Study. The analysis explores varied modes of care that exist in rural and urban areas, as well as the role of internal migration in influencing methods of old-age support. The results paint a rich description of the contemporary dynamics between filial obligations and economic development.

Keywords Aging · Family support · Well-being · China · Intergenerational transfers

Introduction

Rapid demographic change, economic growth, and evolving social norms have put increasing pressure on the traditional system of filial piety and family-based eldercare in China. As an aging crisis primed to stress both formal and informal safety nets looms on the horizon, the question of how to sustain a graying population has moved from a topic with a long history of scholarly study (e.g. Sankar 1989; Whyte 2003; Cook and Powell 2007) to the forefront of State policy discourse. News from the Communist Party's Third Plenum meeting in November, 2013 included announcements of both raising the retirement age and loosening the one child policy in hopes of forestalling a shrinking working-age population (Xinhua 2013; The Economist 2014).

The crisis is not only one of numbers, but cultural as previously understood filial values compete with increasing demands on adults' time and financial resources. The

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Party has reacted in turn, including reissuing an updated version of the 600 years old text "The 24 Paragons of Filial Piety" to reeducate adult children on responsibilities to their parents (Jacobs and Century 2012), and revising the law on the Protection of Rights and Interests of the Elderly to formalize filial obligations and provide legal standing should one's children shirk their economic and spiritual responsibilities (Wong 2013).

As demographers often note, China may soon grow old before it grows rich. The recent dividends born of an abundance of working-age citizens are soon poised to tilt toward an aged population at a level of per capita resources that places China only 92nd in the world, significantly behind its equally graying neighbors Japan (30th) and the Republic of Korea (37th) (World Bank 2014a). The current favorable ratio of four workers per elderly adult is forecasted to fall to three by 2030 and sink to 1.6 by 2050 (World Bank 2014b). Policymakers have certainly not been blind to these challenges and are at work redesigning formal social care. However, while efforts to enroll citizens in new initiatives such as the Rural Pension Program have resulted in a large uptake in coverage, it's questionable whether an equitable formal safety net is sustainable under long term obligations or short term stress – 14 out of 32 provinces ran deficits in their pension funds in 2011 (Zuo 2013), and rural pensions average payouts only 44 % of the poverty line compared to nearly 800 % in urban areas (Zheng et al. 2013).[1]

It is widely understood that informal family support can be a stabilizing force in settings where formal social safety nets are either missing or contain gaps. This is particularly true in China, where cultural norms obligate children to aid their aging parents, and transfers can make up over 20 % of household resources for the elderly (Park et al. 2012). At the same time, this cultural norm is increasingly at odds with competing demands on adult children's financial resources in an increasingly utilitarian society (Li et al. 2005; Ikels 2008). Despite the magnitude and significance of intergenerational transfers and their importance to both science and policy, little is known empirically at the national level concerning the current determinants of child to parent transfers. This paper benefits from the use of new, nationally representative data from the 2011–12 China Health and Retirement Longitudinal Study (CHARLS) to overcome previously existing data limitations. Using multivariate regression analysis and drawing on a sample of approximately 5000 elderly households across the country, this analysis illustrates the correlates of receiving monetary and in-kind transfers and updates a subset of results in Lei et al. (2012) that examined Gansu and Zhejiang provinces with data from the 2008 CHARLS pilot.[2]

Throughout the analysis we pay particular attention to a divergence in the experience of rural versus urban elderly that has been exacerbated by the breakup of rural collectives, pro-market reforms, rapid migration, urbanization, and a widening wealth gap (Li et al. 2005; Zhang and Kanbur 2005; Zhang 2009). While approximately 75 % of older adults in urban areas have retired and face a position of relative privilege, more than half of rural elderly remain employed (Zheng et al. 2013). This paper serves as a nationally representative view of this phenomenon to complement a significant body of qualitative and localized research (e.g. Yan 2003; Zhang and Wang 2010).

[1] Based on the official poverty threshold of 2300 yuan.

[2] Lei et al. (2012) uses a sample of 800 households from the 2008 CHARLS pilot survey collected in Gansu and Zhejiang provinces in preparation for the 2011 national baseline analyzed in this paper.

The results of the analysis reveal a number of clear patterns in the data. We show that transfers among family members are closely tied to their living arrangements, including whether an elderly parent is co-residing with any of their children or has a child that is a migrant. Given the complex costs and benefits facing potential migrants, transfer obligations to the elderly may be a particularly important driver of migration decisions (e.g. Giles and Mu 2007). Additional socioeconomic, demographic, physical, and mental health markers suggest that transfers flow from children toward those parents most in need, updating and extending past work such as Lee and Xiao (1998) to include a richer set of parental characteristics.

The following section briefly discusses the formal pension system, economic approaches to studying informal family support, and previous empirical evidence on intergenerational transfers in China. Section 3 then outlines the CHARLS data set and the multivariate regression approach, while Section 4 presents empirical findings. The results paint a rich description of the contemporary dynamics between filial obligations, economic development, and challenges on the horizon.

Elder Support and the Lack of a National Social Safety Net in China

Long literatures in the social sciences have studied intergenerational exchange and transfers between family members. The Chinese context has provided particular insights given the cultural heritage and norms surrounding filial piety and the expectations of "upstream" transfers to support parents during old-age. However, the traditional system is increasingly at odds with accelerating population aging, a reduction in multigenerational co-residence, drastic differences in local development, and the rise of fee based eldercare to substitute for intra-family assistance (Zhang 2009).

The current Chinese pensions system has its roots in non-contributory legacy pensions paid by state-owned enterprises as part of Mao's "iron rice bowl" (Leung 1994). Starting in 1997, these pensions transformed into the contribution-based Urban Enterprise Pension System which now counts approximately 300 m urban residents among its enrollees (Pozen 2013). As primarily self-employed agricultural workers, rural residents had no such safety net prior to the 2009 rollout of the Rural Pension scheme in which local governments' subsidize voluntary contributions made to individual accounts. Coverage under the rural scheme has grown to near 500 m participants in only 5 years (Pozen 2013), although only providing a level of support well below the poverty line. A similar plan is available for non-employed urban residents, while civil-service pensions requiring no mandatory contributions remain available for a select few.

As many have noted, the existing pension programs breed inequality, with the Economist (2012) going as far as describing the system as a "kind of apartheid" due to highly unequal payouts across and within urban and rural areas. Guarantees range from a low of 55 yuan per month in rural areas to upwards of 2000 yuan per month in some urban, coastal areas (Shen and Williamson 2010). Decentralized administration of the funds by provincial and local governments has not only kept welfare levels uneven, but provided opportunities to misappropriate funds from workers' accounts to pay for today's pension obligations as well as withdraw funds for uses in housing and infrastructure expansion (Pozen 2013; Zheng et al. 2013). While a system of current

contributions supporting existing obligations may appear sustainable in the short term, quickly changing demographics will make a pay-as-you-go arrangement exceedingly difficult to maintain in the future.

In an environment of unstable and insufficient social safety, family transfers remain a primary source for support for many elderly, particularly those in rural areas. Table 1 shows this pattern in the CHARLS data when respondents over the age of 60 with at least one child were asked about their primary source for old-age support.

While children (69.8 %) and pensions (22.7 %) are the two primary support sources in the full combined sample, their importance is drastically different in rural and urban areas. In urban locations, only a third of respondents rely on their children, while near two-thirds see pensions as their primary means of support. Rural residents are not as fortunate, with over 80 % looking to their adult children to sustain themselves in old-age.[3] Savings, transfers from other relatives besides children, and commercial annuities make up a remaining small fraction of support systems.

It follows that economists have come to consider the family as a particularly important institution. Altruistic motives and repeated interactions with relatives can help families achieve the best possible outcomes for their members (e.g. Becker 1981; Chiappori 1988, 1992). This is particularly true for the extended family, which while often dispersed across households, maintains connections to the larger unit. Numerous studies have illustrated the economic importance of extended family networks including Altonji et al. (1992), Hayashi et al. (1996), and Cox and Fafchamps (2007) among many others. The family can be an effective way to spread risk (Foster and Rosenzweig 2001), and intrafamily transfers may be used to provide an informal safety net in times of need (e.g. Cochrane 1991; Genoni 2012; Dalton and LaFave 2015).

Often motivated by norms and expectations surrounding Confucian notions of filial piety, China has provided a particularly important setting for research on extended families and intergenerational exchange. Transfers from adult children to elderly parents are known to be a means of informal insurance for low-income individuals in urban areas (Cai et al. 2006), and to increase in magnitude when needed to cover large medical expenses (Wu and Li 2014). Other evidence notes that transfers may be a form of repayment for previous investments by parents, or as a way for migrants to compensate parents for providing child care to those left behind (e.g. Secondi 1997; Giles et al. 2010; Cong and Silverstein 2011). Transfers in the Chinese context are comprised not only of monetary, but also in-kind gifts such as food (Zhang and Wang 2010), and frequently coincide with lump sums at Spring Festival rather than continuous gifts throughout the year. Distinguishing between varied modes and frequencies of transfers is a key part of the analysis below.

Given rapidly changing economic development, population aging, and stresses on the existing pension system, the incidence and determinants of intergenerational transfers is a particularly relevant and important topic in contemporary China. However, as noted in Lei et al. (2012), studying intergenerational exchange between adult children and their parents has been particularly difficult with previous Chinese data. Of particular concern is that the majority of studies are not able to identify the source of a transfer – i.e. whether or not an elderly

[3] Note that these numbers mask the discrepancy of what "pension" means in terms of expected financial support to an urban versus rural resident.

Table 1 Primary source for old-age support

	Full Sample (1)	Urban (2)	Rural (3)
Percent of Sample Responding […]			
Children	69.8	33.7	81.1
Pension or Retirement Salary	22.7	59.5	11.2
Savings	2.5	2.8	2.4
Other	7.5	6.8	7.8

Values represent the percentage of the sample that responded each source was their primary source of old-age support. Full sample $N = 4994$, urban $N = 1815$, rural $N = 3809$

individual is receiving a transfer from their children or another relative – or are often geographically limited to a particular rural or urban areas. CHARLS, a nationally representative, ongoing longitudinal survey of those 45 and older based on the Health and Retirement Study (HRS) in the United States, alleviates these concerns and provides detailed nationally representative data on transfers and a rich set of additional variables.

CHARLS Data and Empirical Methods

This paper draws on the first national wave of CHARLS collected in 2011–12. The survey includes detailed information on individuals, households, and the communities in which they live for approximately 10,000 households in 28 provinces. The target respondents are a randomly selected subgroup of the population with sampling done at the county, neighborhood, and household level (see Zhao et al. 2013 for a further description). Among many detailed questions contained in the survey are modules on biomarkers, health status, cognition, income, family support, and demographic markers.

As this paper focuses on financial support elderly parents receive from their adult children, we utilize the approximately 5000 households where the primary respondent is at least 60 years of age and has one or more children. These households are dispersed throughout the country with approximately 75 % living in rural areas, and include parents who co-reside with their children as well as those with non co-resident children.

In order to collect information on financial assistance, each respondent is asked about economic help they and their spouse gave and received from non co-resident family members in the previous year.[4] While questions on connections to parents, children, grandchildren, and other relatives are included, this paper focuses on the traditional child to parent transfer, where the unit of analysis is a household. The structure of the survey allows one to delineate transfers between direct monetary support and in-kind gifts, as well as whether they were received on a regular basis or irregularly throughout the previous year.[5]

[4] This implies transfers within the household between adults and their co-resident children are beyond the scope of this analysis.

[5] Note that "in-kind" refers to physical goods such as grain and cooking oil and does not include time transfers.

Table 2 documents the support that the sample households received from their children in the previous year. Column 1, including both rural and urban households, indicates that slightly under half of all households receive some sort of financial transfer from their children. Between the different modes, direct monetary support remains more common than in-kind transfers, as does irregular exchanges compared to financial support on a monthly, or regular schedule.[6]

Columns 2 and 3 emphasize how transfer patterns differ by substantial margins across rural and urban areas – a distinction we focus on throughout the remainder of the paper. This may be indicative of a number of factors including changing social norms in urban areas, a smaller number of children for urban elderly to call upon, or that formal support available to those in urban areas crowds out the informal, family safety net. Further analysis in the next section uses multivariate regressions to assess the sensitivity of the mean patterns in Table 2 to varying characteristics of parents and their children.

Table 3 provides descriptive statistics for additional variables used in the analysis. Of particular note are the household composition and co-residence patterns indicated at the top of the table, with approximately 45 % of parents co-residing with at least one adult child in both urban and rural localities. The sample mean for number of co-residing children conditional on having one or more is 1.2 for both rural and urban households. Thus while co-residence is common, co-residence with more than one child is not common.

Rural parents have slightly larger families on average compared to urban residents, 3.47 to 2.78 children, and each are fairly likely to have one migrant child, where a migrant is defined as a child living outside of the parents' village or neighborhood whose hukou location does not match their place of residence.[7] As migrant children carry a strong financial link to their parents (e.g. Cong and Silverstein 2011), this is a particularly important aspect of family composition to understand as it relates to old-age support.

Given that the primary respondent in the household is randomly selected between the spouses, Table 3 shows the sample is relatively balanced between male and female respondents. Approximately 30 % are already widowed (45 % of women and 16 % of men), and education varies drastically across the sample. Given a substantial fraction of this sample came of age during the Great Leap Forward and Cultural Revolution, educational attainment reflects both limited resources and policy-induced lapses in schooling.

This paper also makes use of detailed cognitive, psychosocial, and physical health measures collected in the study. We include an indicator variable for knowing the correct date as a marker of cognition, as well as one's score from the Center for Epidemiologic Studies Depression Scale ten-question survey (CESD-10).[8] Scores of 10 or above, which characterizes approximately 40 % of the sample, are indicative of depressive symptoms (Andresen et al. 1994). The depressive burden is unequally distributed across gender and location, with 49 % of women falling in the depressive

[6] CHARLS defines irregular transfers as those given at Spring Festival, and/or Mid-Autumn Festival, birthdays, weddings, funerals, or other idiosyncratic times.

[7] Alternative definitions of "migrants" based on differing distances from the parents produce consistent results throughout.

[8] These include agreeing or disagreeing with statements such as "I was bothered by things that don't usually bother me."

Table 2 Receipt of transfers from children

	Full Sample (1)	Urban (2)	Rural (3)
Percent of Sample Receiving […]			
Any transfer	48.5	33.8	53.0
Monetary transfer	43.0	29.3	47.3
Regular	9.9	7.5	10.7
Irregular	35.4	23.2	39.1
In-kind transfer	19.8	13.4	21.8
Regular	2.2	0.8	2.6
Irregular	18.1	12.8	19.7

Values represent the percentage of the sample that received each type of transfer. Full sample $N = 4994$, urban $N = 1815$, rural $N = 3809$

range compared to 34 % of men, and 44 % of rural residents compared to 29 % of those in urban areas.

Respondents are also asked a battery of activities of daily living (ADLs) questions that are useful markers of physical function and health (see Strauss and Thomas 2007). For the purposes of this study, we follow Lei et al. (2012) and include an indicator for whether the individual reports difficulty with any of the activities – running or walking 1 km, getting up from a chair, climbing stairs, stooping, extending one's arms, carrying a moderate amount of weight, or picking up a coin from a table. Consistent with other aging studies, over half of the population is limited by this definition, driven primarily by reported difficulty running 1 km.[9] Health status is also captured by whether the respondent has been diagnosed with arthritis or heart disease, two common ailments in the population.[10]

Empirical Regression Approach

The rich, cross-sectional data makes it possible to conduct a descriptive analysis of transfer patterns amongst the elderly throughout China. The regression approach described below aims to illustrate characteristics that are predictive of a parent receiving a regular or irregular transfer from their children. This is done with the following linear probability regression model:[11]

$$I(T_{ic} = 1) = \beta X_{ic} + \alpha_c + \varepsilon_{ic} \quad (1)$$

where the left-hand side dependent variable is a binary indicator equal to 1 only if individual i in county c received a transfer, $T_{ic} = 1$, from any of their children in the past year. The right hand side variables include individual and family characteristics

[9] Excluding running 1 km from the definition of the ADL marker does not change the regression results that follow.

[10] Heart disease includes heart attacks, coronary heart disease, angina, congestive heart failure, or additional heart problems.

[11] The results are consistent with using a conditional-logit model to constrain the predicted probability of receipt of transfers between zero and one.

Table 3 Descriptive statistics

	Full Sample (1)	Urban (2)	Rural (3)
Co-recide with children (%)	46.8	44.4	47.5
	(0.71)	(1.44)	(0.81)
N. co-resident children	0.56	0.52	0.57
	(0.01)	(0.02)	(0.01)
N. non co-resident children	2.75	2.26	2.90
	(0.02)	(0.04)	(0.03)
Any migrant children (%)	35.8	27.6	38.4
	(0.68)	(1.30)	(0.79)
Age	69.2	69.2	69.2
	(0.11)	(0.23)	(0.13)
Female (%)	49.2	53.6	47.8
	(0.71)	(1.45)	(0.81)
Married (%)	67.4	68.9	66.9
	(0.66)	(1.34)	(0.76)
Widowed (%)	30.4	28.6	31.0
	(0.65)	(1.31)	(0.75)
Owns residence (%)	81.3	79.7	81.7
	(0.56)	(1.18)	(0.63)
Attained Education (%)			
Illiterate	35.7	19.2	40.9
	(0.68)	(1.14)	(0.80)
Middle school and below	56.8	57.6	56.5
	(0.70)	(1.44)	(0.80)
High school and above	7.5	23.2	2.7
	(0.37)	(1.23)	(0.26)
Knows correct date (%)	33.4	60.0	25.2
	(0.67)	(1.42)	(0.70)
CES-D depression score	9.11	7.16	9.72
	(0.09)	(0.17)	(0.11)
Any ADL limitations (%)	55.5	50.0	57.2
	(0.50)	(1.00)	(0.58)
Arthritis (%)	37.2	30.9	39.1
	(0.68)	(1.34)	(0.79)
Heart disease (%)	15.7	24.7	12.9
	(0.15)	(1.25)	(0.54)
N. Obs	4994	1185	3809

Table reports means and standard errors for variables of interest. Sample consists of respondents age 60 and above with at least one child

described in Table 3, X_{ic}, and county fixed-effects indicators, α_c, to capture common, average characteristics shared by individuals in their communities. Given the stark geographic differences in infrastructure and availability of services, prior work using CHARLS data has acknowledged the importance of the inclusion of county fixed effects in such models (e.g. Lei et al. 2013). Robust standard errors are calculated allowing for clustering at the county level.

The coefficients of interest, β, relate how changes in variables included in X, such as an increase in the number of children, relate to changes in the probability of receiving a transfer. The next section describes the results of this analysis for the full sample, and urban and rural households separately, as well as differences between determinants of monetary and in-kind support.

Findings and Discussion

Before examining regression results, Figs. 1 and 2 display two possible determinants of transfer receipt to visually represent the multivariate regression approach.[12] Figure 1 plots the probability of receiving transfers against age of the respondent for the full sample and disaggregated by rural and urban regions. The data shows that elderly respondents are more likely to have received a transfer in the past year, and that transfers are more common in rural areas at every age. As the sample is seventy five percent rural, the full-sample average closely tracks the rural group.

Similarly, Fig. 2 illustrates that the probability of receiving a transfer is increasing in the number of children the respondent has in both rural and urban areas. The positive slopes visible in both figures reflects that there is a margin to intergenerational transfers – were child to parent transfers universal, the plots would be flat lines with no margin of interest to examine.

The advantage of moving to multivariate regression analysis is the ability to simultaneously examine multiple determinants of transfers rather than single covariates. Tables 4 and 5 present coefficient estimates from Eq. (1) examining how the predicted probability of receiving a transfer changes with family composition, the type of transfer, and parental characteristics. Table 4 begins with an analysis of the full sample, with columns 1 through 3 noting the marginal changes from the multivariate linear probability model in relation to receiving any – monetary or in-kind – transfers from a child.

The results suggest statistically robust patterns of exchange exist that are closely tied to family composition. As seen in Fig. 2, column 1 illustrates that families with more children have increased likelihoods of receiving a transfer – conditional on holding the other characteristics in the model fixed, one additional child raises the predicted probability of receipt by 6.7 percentage points. This is over a 13 % increase given the baseline mean of 48.5 %.

However, if families choose to sort their members across physical distance and space as has been suggested in other contexts (Hamoudi and Thomas 2014), it is likely that the effect on transfer receipt is different for co-resident versus non co-resident children. Column 2 disaggregates the main effect into the marginal change due to an additional child living with the respondent parent and an additional child living outside of the parents' residence. The results show having an additional co-resident child reduces the probability of receiving a transfer by 5.2 percentage points, while an additional child living apart has a positive 6.7 percentage point effect. That the two are nearly directly opposite each other suggests co-resident and non co-resident children may act as substitutes, either exchanging time transfers through co-residence or financial support.

[12] Figures 1 and 2 are nonparametric plots generated with local Epanechnikov kernel smoothing using a rule-of-thumb bandwidth. Alternative kernels or bandwidth produce figures consistent with those shown here.

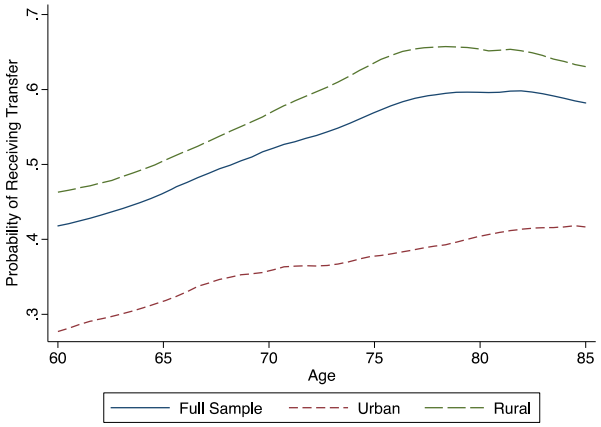


Fig. 1 Age profile of the probability of receiving transfers

Column 3 further disaggregates the non co-resident children into migrants and non-migrants. Consistent with evidence in Giles et al. (2010), the disaggregation reveals migrant children are significantly more likely to send financial support to their parents than children living close by, as an additional migrant child increases the probability of receipt by 10.2 percentage points relative to 6.1 for non-migrants. These findings provide evidence of a complex intrafamily decision making process linking transfer motives, living arrangements, and a distribution of elder-care responsibilities among adult siblings. This is consistent with evidence from economics highlighting intrahousehold bargaining within Chinese families (Li and Wu 2011; Wang 2014), as well as work from gerontology on adult children coordinating their parents' care (e.g. Zhang and Wang 2010).

Additional socioeconomic status variables included in the regressions suggest those parents most at risk or in-need are more likely to receive help from their children. This is reflected by an increase in the likelihood of receiving transfers for those living in rural areas, widows, and those who do not own their own

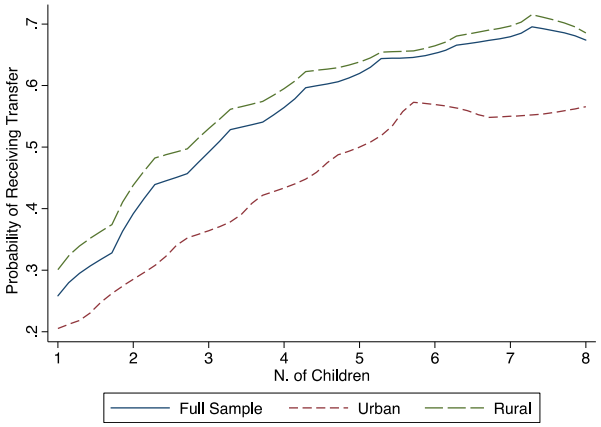


Fig. 2 Probability of receiving transfers increases with the number of children

Table 4 Probability of receiving transfers

	Dependent variable: I (Received […] Transfer from Children)				
	Any			Monetary	In-Kind
	(1)	(2)	(3)	(4)	(5)
Number of […] children					
total	0.067*** (0.006)				
co- resident		-0.052***(0.011)	-0.047***(0.011)	-0.038***(0.011)	-0.014*(0.008)
non co- resident		0.067*** (0.005)			
non co-res non--migrant			0.061*** (0.005)	0.062*** (0.005)	0.030*** (0.005)
non co-res mi-grants			0.102*** (0.009)	0.105*** (0.009)	0.041*** (0.008)
Age (60–69 omitted)					
70–79	0.024 (0.020)	0.001 (0.020)	0.007 (0.019)	−0.000 (0.018)	0.028* (0.014)
80+	0.011 (0.027)	−0.008 (0.027)	0.002 (0.027)	−0.010 (0.026)	0.036 (0.024)
Female	0.021 (0.015)	0.021 (0.015)	0.022 (0.015)	0.020 (0.015)	0.019 (0.012)
Urban	−0.045* (0.025)	−0.048* (0.025)	−0.043* (0.025)	−0.043* (0.027)	−0.037* (0.020)
Widowed	0.053*** (0.016)	0.064*** (0.016)	0.067*** (0.016)	0.067*** (0.015)	0.024* (0.014)
Owns residence	−0.072*** (0.020)	−0.037*** (0.020)	−0.032*** (0.021)	−0.031*** (0.019)	−0.048*** (0.018)
Education (illiterate omitted)					
Middle school or below	0.005 (0.017)	0.004 (0.017)	0.002 (0.017)	0.011 (0.016)	−0.007 (0.015)
High school and above	−0.030 (0.030)	−0.032 (0.029)	−0.034 (0.029)	−0.031 (0.028)	0.001 (0.026)
Knows correct date	−0.018 (0.021)	−0.020 (0.020)	−0.020 (0.020)	−0.020 (0.018)	−0.019 (0.015)
CES-D score	−0.003** (0.001)	−0.003*** (0.001)	−0.003*** (0.001)	−0.003*** (0.001)	−0.002* (0.001)
Any ADL limitations	0.011 (0.016)	0.009 (0.016)	0.010 (0.016)	0.007 (0.015)	−0.010 (0.012)
Arthritis	0.032** (0.015)	0.031** (0.015)	0.031** (0.015)	0.015 (0.014)	0.020 (0.013)
Heart disease	0.058*** (0.018)	0.061*** (0.018)	0.061*** (0.018)	0.043*** (0.019)	0.042*** (0.016)
N. observations	4994	4994	4994	4994	4994

Table reports coefficients and standard errors (in parenthesis) for linear probaility models of the receift of transfers. All regressions include county fixed effects, sampling weights, and standard errors clustered at the county level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Table 5 Probability of receiving transfers -urban and rural samples

	Dependent variable: I (Received […] Transfer from Children)					
	Urban			Rural		
	Any (1)	Monetary (2)	In-Kind (3)	Any (4)	Monetary (5)	In-Kind (6)
Number of […] children						
co-resident	−0.014	0.003	0.000	−0.056***	−0.046***	−0.014
	(0.033)	(0.032)	(0.023)	(0.014)	(0.013)	(0.009)
non co-res non-migrant	0.078***	0.078***	0.028***	0.062***	0.064***	0.032***
	(0.015)	(0.014)	(0.010)	(0.006)	(0.006)	(0.005)
non co-res migrants	0.151***	0.148***	0.084*	0.099***	0.102***	0.044***
	(0.036)	(0.037)	(0.049)	(0.010)	(0.011)	(0.010)
Age (60–69 omitted)						
70–79	−0.036	−0.061	0.005	0.004	0.002	0.034*
	(0.046)	(0.043)	(0.037)	(0.023)	(0.021)	(0.018)
80+	−0.021	−0060	0.076	−0.003	−0.014	0.046
	(0.018)	(0.064)	(0.066)	(0.030)	(0.031)	(0.032)
Female	0.061*	0.067**	0.009	0.006	0.006	0.007
	(0.034)	(0.031)	(0.033)	(0.020)	(0.020)	(0.014)
Widowed	0.042	0.028	0.037	0.086***	0.090***	0.025
	(0.048)	(0.043)	(0.033)	(0.020)	(0.019)	(0.018)
Owns residence	−0.154**	−0.132*	−0.107***	−0.022	−0.024	−0.048**
	(0.074)	(0.074)	(0.039)	(0.026)	(0.024)	(0.023)
Education (illiterate omitted)						
Middle school or below	−0.013	0.004	0.038	−0.008	−0.001	−0.011
	(0.046)	(0.044)	(0.036)	(0.023)	(0.019)	(0.018)
High school and above	−0.010	−0.004	−0.035	−0.116*	−0.119**	−0.021
	(0.058)	(0.058)	(0.032)	(0.060)	(0.047)	(0.055)
Knows correct date	−0.048	−0.035	−0.005	0.006	−0.006	−0.003
	(0.048)	(0.050)	(0.038)	(0.025)	(0.021)	(0.022)
CES-D score	0.004	0.004	0.005**	−0.004***	−0.004***	−0.002*
	(0.003)	(0.004)	(0.002)	(0.001)	(0.001)	(0.001)
Any ADL limitations	−0.050	−0.041	−0.066**	−0.010	−0.015	−0.005
	(0.042)	(0.042)	(0.027)	(0.018)	(0.020)	(0.016)
Arthritis	−0.013	−0.030	0.038	0.045**	0.023	0.037**
	(0.043)	(0.036)	(0.048)	(0.019)	(0.018)	(0.017)
Heart disease	0.102***	0.063**	0.046	0.063**	0.058**	0.044**
	(0.034)	(0.031)	(0.034)	(0.027)	(0.028)	(0.022)
N. observations	1185	1185	1185	3809	3809	3809

Sample in columns 1–3 consists of only urban households, and rural households in columns 4–6. Table reports coefficients and standards errors (in parenthesis) for linear probility models of the receipt of transfers. All regressions include county fixed effects, sampling weights, and standard errors clustered at the county level. $^{**}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$

residence. Health outcomes are also strongly predictive of transfer receipt, as being diagnosed with arthritis increases the probability of receipt by

approximately 3 percentage points and heart disease relates to an increase of 6 percentage points.

A notable exception to this trend of transfers flowing to vulnerable parents is the depressive symptoms result. Although small in magnitude, the negative coefficient of −0.003 suggests the more depressed a parent is the *less* likely they are to receive transfers. The magnitude reflects an approximate 3-percentage point decline in the predicted likelihood of receiving transfers for falling into the "depressed" range of CESD-10 scores. Models with interactions between gender and CESD-scores available upon request show that the impact of depression is larger for women than men, suggesting depressed women are particularly at risk. This is consistent with the findings in Lei et al. (2012) from the 2008 CHARLS pilot sample, as well as prior work on stigma and neglect amongst depressed elderly (Dong et al. 2008; Wang and Zhao 2012). Somewhat surprisingly, age, education, and cognition as captured by date recognition are statistically unrelated to receipt once other characteristics are accounted for.

Columns 4 and 5 examine whether the patterns differ for monetary transfers as opposed to in-kind support. The dependent variable in each of the regressions is changed to equal one only when the specific type of transfer is received and zero otherwise. While Table 2 illustrated there was a mean difference in the receipt of each type – monetary transfers being far more common – this analysis helps to understand whether the determinants of each type of transfer differ.

The sign and magnitudes of the coefficients in columns 4 and 5 are relatively consistent between the two modes with a few notable exceptions. First, the additional increase in transfer receipt due to migrant children seen in column 3 is concentrated in monetary rather than in-kind transfers. While having an additional migrant child increases the predicted likelihood of receiving both types of transfers, the marginal change is greater for cash transfers, a 10.5 percentage point increase, relative to a 4.1 percentage point increase for in-kind support.

Of the remaining covariates, both types of transfers are particularly responsive to heart disease and only the coefficient on widowed is statistically different between the two columns, with widowhood inducing a higher likelihood of monetary rather than in-kind support.

Urban Versus Rural Differences

Given the stark differences in mean transfer receipt between those households located in urban and rural settings visible in the summary statistics and illustrative figures, Table 5 displays coefficients from Eq. (1) estimated separately for the urban and rural samples. The specification disaggregating children into co-resident, migrant, and non-migrants is maintained throughout the regressions.

Differences are immediately visible in the relationship between additional co-resident children and the likelihood of receiving a transfer. While the rural sample reflects the pattern shown in Table 4 where co-resident and non co-resident children offset, there is no statistically significant negative effect of additional co-resident children for urban households. Margins are similarly spread for migrant versus non-migrant children, although migrants from urban

parents seem particularly likely to remit monetary transfers home. Having a migrant child raises urban parents' likelihood of receiving transfers by 15.1 percentage points (column 1) compared to 9.9 percentage points for rural parents (column 4). This difference is potentially reflective of opportunities available to urban migrants compared to those with rural status.

Other notable differences between the two samples appear in the coefficients on gender, marital status, owning a residence, and education. Urban women appear to have elevated probability of receipt, as do rural widows. Moreover, those individuals in rural areas who attained high school, only 2.7 percent of the rural sample, are approximately 12 percentage points less likely to receive a transfer than an illiterate individual (the omitted reference category). As with other covariates such as heart disease, this is likely reflective of transfers flowing to those in need.

Conclusion

The results presented above serve to describe the current state of child to parent transfers using new, nationally representative survey data. Clear patterns emerge regarding residency arrangements, migration, health, and socioeconomic status in both rural and urban areas. Consistent with models of family members providing an informal social safety net, transfers are closely related to family composition and living arrangements, and appear to flow to parents of low means and those in poor health.

As in other global contexts, co-residence and financial support are not only indicative of the importance that children play in contemporaneously insuring the well-being of the elderly, but also of connections that persist over the life-course between family members. What many policy makers now fear are changes on the horizon that threaten the stability of this current system. These include evolving cultural expectations of what constitutes support, rising dependency, and shrinking family size – a particularly severe concern given the strong positive connection between the number of children and receipt of transfers. The patterns described above, and the State's ability to provide a suitable social safety net, will certainly look different in decades to come given declining fertility rates that rest at 1.7 today (World Bank 2014c) compared with the mean number of children in the sample of 3.3.

These changes present a number of challenges to intrafamily relationships as well as continued economic growth. With the Party recently revising annual growth expectations down to 7.5 % from a 10 year average of 10.4 % (Xinhua 2014; World Bank 2014d), even obtaining a relatively modest growth rate by Chinese standards appears to be a challenge given aging-linked concerns such as labor shortages (Zhang et al. 2011), declining agricultural productivity (Li and Sicular 2013), and increasing financial obligations required to support a graying population. While the results suggest family transfers remain at the forefront of old-age support, the stability of the system depends on the interactions of elderly families and government social policy. As the economy, norms, and policies continue to evolve, so too must research to continue to understand the dynamics of old-age support in a rapidly changing environment.

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Compliance with Ethical Standards

Conflict of Interest The author declares that there is no conflict of interest.

Informed Consent As there is no person or personal data appearing in the paper, there is no one from whom a permission should be obtained in order to publish personal data.

Ethical Treatment of Experimental Subjects (Animal and Human) All procedures performed in studies involving human participants were in accordance with the ethical standards of the institutional and/or national research committee and with the 1964 Helsinki declaration and its later amendments or comparable ethical standards.

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